

Exploratory Data Analysis (EDA) Report

Internship OIBSIP – Third Project

Idea: Unveiling the Android App Market: Analyzing Google Play Store Data

Dataset: apps.csv + user_reviews.csv (SQL-only)

Introduction

This report presents a structured analysis of Google Play Store data.

The aim is to clean, categorize, and interpret app and user review information to uncover market dynamics, user engagement patterns, pricing strategies, and sentiment trends.

The findings provide actionable insights for developers, businesses, and evaluators.

Category Exploration

- FAMILY has the largest app count (1827), but GAME dominates reviews (10,640).
- Medical and Health apps show high engagement relative to app count.
- AUTO_AND_VEHICLES and EDUCATION categories lead with strong ratings (~4.4+) and balanced sentiment.
- Free apps dominate sentiment volume (31,504 Neutral, 23,671 Positive), while Paid apps are fewer but generally positive.

Explanation: Categories reveal where the app ecosystem is concentrated and how users interact with different domains.

Games attract attention but also polarizing feedback, while Education and Health apps show consistent satisfaction.

Metrics Analysis

- Popular creative apps like "Sketch - Draw & Paint" (50M installs, rating 4.5) and "Ibis Paint X" (10M installs, rating 4.6) dominate installs.
- Top reviewed apps include "Photo Editor & Candy Camera" (400 reviews), "CBS Sports App", and "Angry Birds Classic".
- Finance (140.39) and Lifestyle (124.26) categories have the highest average prices, while Tools and Communication remain low-cost.
- Larger creative apps maintain strong ratings, but smaller apps can also achieve high satisfaction (Photo Designer, 5.5 MB, rating 4.7).
- FAMILY and MEDICAL categories show significant paid app presence, while Games remain predominantly free.

Explanation: Metrics highlight how installs, reviews, pricing, and app size influence user perception.

High installs often correlate with strong ratings, but niche apps can achieve satisfaction despite smaller scale.

Sentiment Analysis

- Overall sentiment: Neutral reviews dominate (32,030), but positives (23,996) outweigh negatives (8,269).
- Games attract the most reviews but also high negativity (2408). Health & Fitness apps show strong positivity (1754 Positive vs 257 Negative).
- Educational and productivity apps trend towards higher polarity (Homework, Google Slides), while entertainment apps show mixed tone.
- COMICS and EVENTS categories are perceived positively, while Games are polarizing.
- High ratings generally align with positive reviews (e.g., Down Dog Yoga, rating 4.9, 40 Positive, 0 Negative). Popular games show polarity between user groups.
- High installs do not guarantee high review counts; engagement varies by app type (Duolingo vs Google Photos).

Explanation: Sentiment analysis reveals user satisfaction levels and emotional tone.

Neutral reviews dominate, but positivity is strong in Education and Health apps. Games, despite popularity, face criticism.

Recommendations

- Focus on categories with high ratings and balanced sentiment (Education, Health & Fitness, Auto).
- Address negativity in Games by improving stability, user experience, and support.
- Explore monetization opportunities in Finance and Lifestyle categories where users accept higher pricing.
- Encourage lightweight apps with strong ratings as they show high satisfaction despite smaller size.
- Leverage sentiment polarity insights to design better user engagement strategies.

Conclusion

The analysis reveals that while Games dominate in volume and reviews, they face sentiment challenges.

Education, Health & Fitness, and niche categories show strong positivity and user satisfaction.

Finance and Lifestyle categories present monetization opportunities.

Overall, the Android app market is diverse, with clear signals for developers and businesses to align product strategies with user expectations.